Enhanced Third-Harmonic Generation in PEDOT:PSS via Femtosecond-Laser-Induced Reduction

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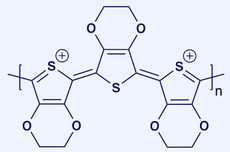
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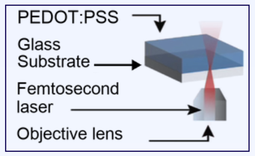
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Introduction & Motivation

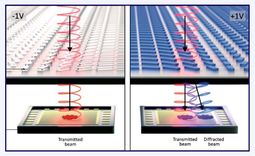
- **PEDOT:PSS** (PH1000) is a conductive and solution-processable polymer whose optical and electrical **properties** can be dynamically **controlled** through its **oxidation state.**
- Its combination of high conductivity, optical transpar-ency and reversible redox activity has enabled various applications in active photonics, bioelectronics, and electrically switchable optical devices.



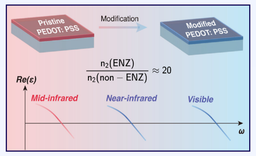
Fs-laser patterning



Electrical tunability



Tunable nonlinearity



1. S. Doshi, *et al.* Adv. Mater. **2025** 37 2415827.
2. S. D. Ludescher, *et al.* Adv. Opt. Mater. **2025** 13 2403271.
3. J. Karst *et al. Science* 374 612-616 **2021**
4. S.. Doshi et al. **2024** Nature Materials 24 205.
5. Q. Hu *et al. ACS Photonics* **2023** 10 10 3612-3620.

Previous works have shown:

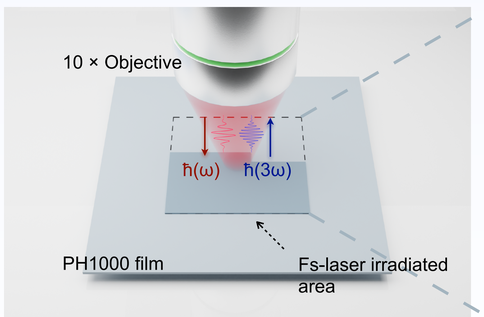
- Fs-laser processing creates stable PEDOT:PSS microstructures.
- Redox control enables active PEDOT:PSS nano-optics.
- Modification changes the nonlinear response of PEDOT:PSS.

Motivation:

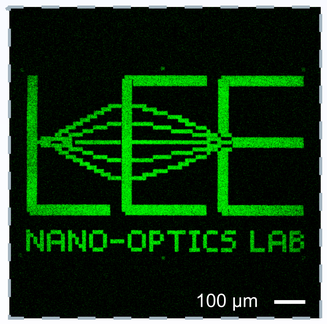
- Investigate whether the direct laser modification of PEDOT:PSS can change its optical properties and be used to dynamically **program** third-harmonic generation (THG).

Experimental Setup

Optical microscopy setup



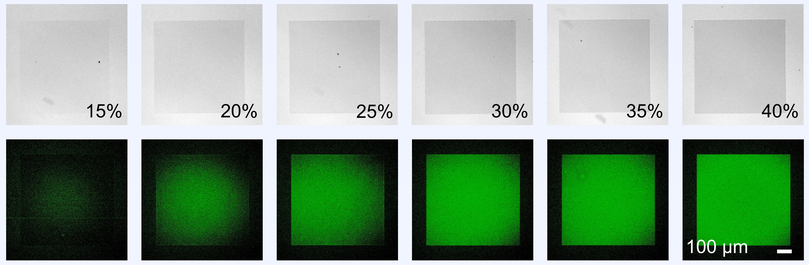
THG pattern



- **Fs-laser irradiation:** 1300 nm, 120-fs, 80 MHz laser pulses focused onto PEDOT:PSS film using a 10× objective.
- **Nonlinear imaging:** THG was collected 433 nm in reflection mode.
- **Spatial control:** laser exposure produced patterned THG contrast.

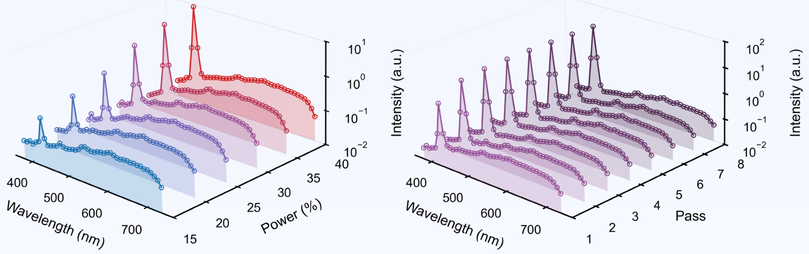
THG Enhancement

Power-dependent modification and THG contrast



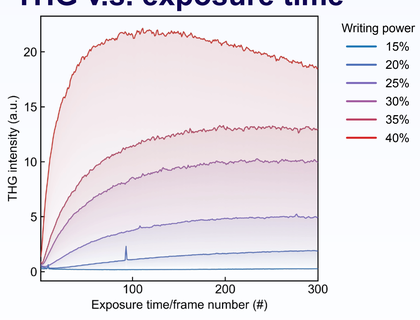
- PEDOT:PSS regions were irradiated from 15% to 40% laser power. THG maps were subsequently acquired at a fixed 15% readout power.
- **Higher writing power leads to stronger THG enhancement.**

THG spectra v.s. power and cumulative exposure

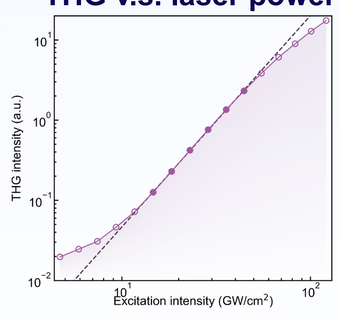


- **Left:** Increasing excitation power from 15% to 40% produces a strong increase in the THG peak near 433 nm.
- **Right:** At a fixed excitation power (35%), repeated spectral passes enhance the THG, revealing a **cumulative laser-induced modification of PEDOT.**

THG v.s. exposure time



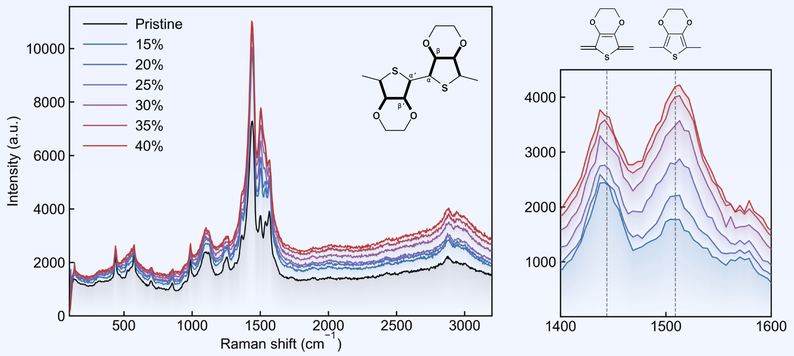
THG v.s. laser power



- **Left:** THG increases cumulatively under repeated fs-laser scanning and approaches a power-dependent saturation level. Higher excitation powers produce faster buildup and stronger enhancement.
- **Right:** The modified PEDOT:PSS film exhibits approximately cubic power scaling, confirming a third-order nonlinear optical process.

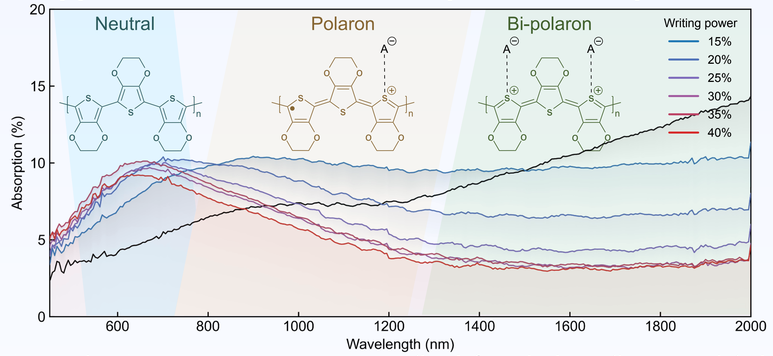
Discussion

Raman signatures of laser-induced de-doping



- Fs-laser exposure increases the PEDOT:PSS backbone modes near 1444 and 1509 cm^{-1} without a measurable peak shift. This spectral evolution is consistent with de-doping and carrier localization in the irradiated regions.

Suppression of infrared absorption after exposure



- Increasing laser exposure suppresses the infrared absorption associated with polaron and bi-polaron states, indicating a reduced doping level.
- The resulting lower absorption at the 1300 nm wavelength reduces optical loss and supports the enhanced THG response.

Outlook & Impact

- **Spatial programmability:** Fs-laser de-doping enables persistent and continuous control of THG.
- **Electrical tunability:** Integration with electrochemical modulation could enable reversible modulation of nonlinear optical responses.
- **Applications:** Nonlinear metasurfaces, optical switching, imaging, and photonic memory.
- **This work:** J. Yan *et al.*, manuscript submitted (2026).

Let's connect!